

The space of *linear dependences* of the vector configuration V is

$$\text{Dep}(V) := \{\mathbf{v} \in \mathbb{R}^n : V\mathbf{v} = \mathbf{0}\} \subseteq \mathbb{R}^n.$$

This is a linear subspace of $\mathbb{R}^n$ of dimension $n - r$. The *signed vectors* of V are given by

$$\mathcal{V}(V) := \{\text{sign}(\mathbf{v}) \in \{+, -, 0\}^n : \mathbf{v} \in \mathbb{R}^n, V\mathbf{v} = \mathbf{0}\} = \text{SIGN}(\text{Dep}(V)),$$

and the *signed circuits* are the signed vectors of minimal nonempty support.

Dually, the space of *value vectors* on a vector configuration, which correspond to linear functions $\mathbf{c} \in (\mathbb{R}^r)^*$, is constructed as

$$\text{Val}(V) := \{\mathbf{c}V : \mathbf{c} \in (\mathbb{R}^r)^*\} \subseteq (\mathbb{R}^n)^*.$$

This is a linear subspace of $(\mathbb{R}^n)^*$ of dimension r . From it, we derive the set of *signed covectors* of the configuration V :

$$\mathcal{V}^*(V) := \{\text{sign}(\mathbf{c}V) : \mathbf{c} \in (\mathbb{R}^r)^*\} = \text{SIGN}(\text{Val}(V)).$$

Here we get the *signed cocircuits* as the signed covectors of minimal nonempty support: they correspond to the linear functions such that the vectors $\mathbf{v} \in V$ which have value 0 linearly span a hyperplane in $\mathbb{R}^r$.

Note that these definitions are consistent with our conventions for the affine case; see Exercise 6.3.

Proposition 6.4. *Let $V \in \mathbb{R}^{r \times n}$ represent a spanning configuration of n vectors in $\mathbb{R}^r$.*

Then $\text{Val}(V)$ is the set of all linear functions that vanish on all the vectors in $\text{Dep}(V)$, and $\text{Dep}(V)$ is the set of all vectors on which all the functions in $\text{Val}(V)$ vanish.

Proof. This follows from the dimension counts $\dim(\text{Dep}(V)) = n - r$, $\dim(\text{Val}(V)) = r$, with the computation $(\mathbf{c}V)\mathbf{v} = \mathbf{c}(V\mathbf{v}) = \mathbf{c}\mathbf{0} = 0$. $\square$

(If we identify $\mathbb{R}^n$ and $(\mathbb{R}^n)^*$ via the standard scalar product, then by the same argument we get that $\text{Dep}(V)$ is the orthogonal complement of $\text{Val}(V)$ in $\mathbb{R}^n$.)

6.3 Oriented Matroids

In this section, we meet oriented matroids for the first time. Just to keep things more exciting (and to make sure that the timid reader isn't overwhelmed by the first sight), we won't lift all the veils on the first encounter. So we won't even define oriented matroids this time, but we promise more for later.

For application to Gale diagrams, it is sufficient to know some basic facts in the case of *realizable* oriented matroids; and we will only do proofs for those below, with the Farkas lemma as a basic ingredient. The general case is similar, but it relies on more notation and on the precise version of the axioms to work with. So the main gain we get from oriented matroid theory for the moment is some notation and terminology (that's where "circuit," "cocircuit," "covector," and so on are from) and the right intuition. Don't underestimate the value of that.

Definition 6.5. Let $V \in \mathbb{R}^{r \times n}$ be a set of vectors that spans $\mathbb{R}^r$.

The *oriented matroid* $\mathcal{M}(V)$ of V is the combinatorial structure encoded by the following four collections of sign vectors:

- the set of circuits of $\mathcal{C}(V)$,
- the set of vectors of $\mathcal{V}(V)$,
- the set of cocircuits of $\mathcal{C}^*(V)$,
- the set of covectors of $\mathcal{V}^*(V)$.

The families of sign vectors arising from a vector configuration V in this way are called a *realizable oriented matroid*.

In the following, we will use small capitals, like $x, u, v \in \{+, -, 0\}^n$ and $c \in (\{+, -, 0\}^n)^*$, to denote (column or row) sign vectors.

For a simple 2-dimensional configuration V we read off as follows:

$$\begin{aligned}
 \text{circuits: } \mathcal{C}(V) &= \left\{ \begin{pmatrix} + \\ - \\ + \end{pmatrix}, \begin{pmatrix} - \\ + \\ - \end{pmatrix} \right\}, & \begin{array}{c} \text{3} \\ \uparrow \\ \text{2} \\ \nearrow \\ \text{1} \end{array} \\
 \text{vectors: } \mathcal{V}(V) &= \left\{ \begin{pmatrix} + \\ - \\ + \end{pmatrix}, \begin{pmatrix} - \\ + \\ - \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}, \\
 \text{cocircuits: } \mathcal{C}^*(V) &= \{(0++), (0--), (+0-), (-0+), (++0), (--0)\} \\
 \text{covectors: } \mathcal{V}^*(V) &= \\
 &= \left\{ \begin{array}{cccccc} (+++), & (---), & (++-), & (-+-), & (+--), & (-++), \\ (0++), & (0--), & (+0-), & (-0+), & (++0), & (--0), \\ (000) \end{array} \right\}
 \end{aligned}$$

In this section, we want to make four main points:

- The data of an oriented matroid are highly structured.
- All four sets of data are equivalent.
- Duality is built into the system.
- The basic constructions of "deletion" and "contraction" are oriented matroid operations that are dual to each other.

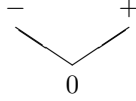
(a) Axiomatics

The collections of sign vectors that make up an oriented matroid are highly structured, and not just random collections. In fact, they arise in the following way.

For any linear subspace $U \subseteq \mathbb{R}^n$, we define the operator SIGN as

$$\text{SIGN}(U) := \{\text{sign}(\mathbf{x}) : \mathbf{x} \in U\} \subseteq \{+, -, 0\}^n.$$

There is a natural *partial order on the set of signs* $\{+, -, 0\}$: we set $0 < +$ and $0 < -$, while $+$ and $-$ are incomparable:



This corresponds to the fact that a number that is slightly perturbed either keeps its sign, unless it is zero, in which case its sign can change to $+$, or to $-$, or remain 0.

On sets of sign vectors $\mathcal{S} \subseteq \{+, -, 0\}^n$, we use componentwise partial ordering: $\mathbf{u} \leq \mathbf{u}'$ if and only if $u_i \leq u'_i$ holds for all positions i . Thus we get, for example,

$$(0+0+000--+-) < (0+-++0--+-)$$

but

$$(0+0+000--+-) \not\leq (00-++0--+-)$$

because of the second position, where $+$ $\not\leq$ 0 .

We use the componentwise partial order to define the operator MIN, which takes all the *minimal nonzero sign vectors* in $\mathcal{S}$:

$$\text{MIN}(\mathcal{S}) := \{\mathbf{u} \in \mathcal{S} \setminus \mathbf{0} : \text{there is no } \mathbf{u}' < \mathbf{u} \text{ with } \mathbf{u}' \in \mathcal{S} \setminus \mathbf{0}\}.$$

We will apply the operator SIGN equally to sets of row vectors and of column vectors. Similarly, we apply MIN both to sets of row sign vectors and of column sign vectors.

With these conventions, we can describe the oriented matroid data for V as follows:

$$\mathcal{V}(V) = \text{SIGN}(\text{Dep}(V)) \quad \mathcal{C}(V) = \text{MIN}(\text{SIGN}(\text{Dep}(V))) = \text{MIN}(\mathcal{V}(V))$$

$$\mathcal{V}^*(V) = \text{SIGN}(\text{Val}(V)) \quad \mathcal{C}^*(V) = \text{MIN}(\text{SIGN}(\text{Val}(V))) = \text{MIN}(\mathcal{V}^*(V))$$

Furthermore, the spaces $\text{Dep}(V)$ and $\text{Val}(V)$ determine each other by Proposition 6.4, so all four sets of data are determined by $U := \text{Val}(V)$. In this sense, we talk about “the oriented matroid $\mathcal{M} = \mathcal{M}(U)$ of the subspace $U \subseteq \mathbb{R}^n$.” The dimension $r := \dim(U)$ is called the *rank* of $\mathcal{M}$. So what we